

Deutsch-CTC is a limit-cycle theory: numerical evidence for cycle solutions, multi-basin dynamics, and a continuum of chronology-coupling regimes

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Abstract

We report three numerical findings that revise the standard fixed-point interpretation of Deutsch’s closed-timelike-curve formulation [1] on small Hilbert spaces. First, the majority of Clifford-structured D-CTC channels do *not* converge to a fixed point: they reach a limit cycle of period k predicted exactly by $\arg(\lambda_2(\mathbf{E})) = 2\pi/k$, where $\mathbf{E}$ is the channel superoperator and λ_2 its second-largest eigenvalue. Cycle solutions have $|\lambda_2| = 1$ marginal modes that preserve input information indefinitely, and their state-distinguisher amplification is $\sim 2\times$ that of the convergent class. Second, the multi-basin structure of these cycles means D-CTC dynamics are *not* ergodic across the full density-matrix space: a single Clifford-class channel admits an average of 25 distinct limits depending on initial state (vs. 1 for Haar-random channels). Third, the previously-reported binary “abstract vs. direct-CTC” distinction in chronology-protection coupling is actually a continuum interpolated by the channel’s direct-CTC content, with the chronology-protection signature appearing gradually above 50% direct encoding. Together these findings imply Deutsch’s self-consistency equation should be re-read as admitting cycle solutions of any period $k \geq 1$, with the Aaronson–Watrous [2] state-distinguisher speedup operating primarily on cycle solutions rather than fixed points. All code, data, and reproduction scripts are at github.com/Zynerji/systrophe (v0.17.0).

1 Introduction and motivation

The Deutsch fixed-point equation for closed-timelike-curve quantum mechanics [1] reads

$$\rho^* = \mathcal{E}(\rho^*) \quad \text{with} \quad \mathcal{E} : \rho \mapsto \text{Tr}_{\text{CR}}[U(\sigma_{\text{CR}} \otimes \rho)U^\dagger] \quad (1)$$

and is universally treated as defining a fixed-point problem. The practical algorithm is to iterate $\rho_{n+1} = \mathcal{E}(\rho_n)$ from a random initial state, stopping when $\|\rho_{n+1} - \rho_n\| < \epsilon$. The Aaronson–Watrous result [2] states that an oracle for ρ^* enables polynomial-time PSPACE computation, including amplifying the distinguishability of two close mixed states beyond the Helstrom bound [4].

In a companion paper [5], we identified the Clifford-structured channel class (permutation $\times$ diagonal-of-fourth-roots unitaries) as the empirical site of Aaronson–Watrous amplification: $\sim 40\%$ of such channels produce purity- > 0.9 fixed points, versus $\sim 0\%$ for Haar.

Three deep-dive experiments documented in this note overturn or sharpen the central assumptions of that picture.

Cycle solutions, not fixed points

We tested 1000 Haar-random unitaries and 1000 Clifford-structured unitaries on ($d_{\text{CR}} = 2, d_{\text{CTC}} = 3$) for *cycle structure*: starting from a random pure initial state ρ_0 , we iterated $\rho_{n+1} = \mathcal{E}(\rho_n)$ and checked whether the trajectory $\{\rho_n\}$ matched a previous iterate within tolerance 10^{-8} within 100 steps.

Result

Table 1: Cycle-length distribution from 1000-channel sample.

Cycle length	Haar	Clifford-like
1 (converged)	65.9%	32.7%
2	17.5%	39.4%
3	5.8%	14.4%
4	0.3%	10.3%
≥ 5 or no cycle in 100 steps	10.5%	0.0%

The majority of Clifford-class D-CTC channels reach a limit cycle, not a fixed point. For Haar channels, fixed-point convergence dominates (66%) but cycle solutions are still present (24% combined).

Spectral prediction

Channels with cycle length k have second-largest eigenvalue $\lambda_2(\mathbf{E})$ on the unit circle at angle $2\pi/k$:

Cycle length	Mean $ \arg \lambda_2 $	Theoretical $2\pi/k$
1 (converged)	0.000	0
2	2.447	$\pi \approx 3.142$
3	2.094	$2\pi/3 \approx 2.094$

The period-3 prediction is essentially exact. The period-2 cases cluster near (but not at) π , suggesting a slight tilt off the negative real axis.

Critically, **period-2 channels have $|\lambda_2| = 1.000$ exactly**, indicating a marginal mode on the unit circle. The iteration’s subdominant mode is neither decaying (which would give convergence) nor growing (which would give instability) — it rotates in place. This marginal mode *preserves input information indefinitely*, which is the structural mechanism for state-distinguisher amplification.

Cycles amplify more than fixed points

Comparing amplification of two close mixed states (input trace distance 0.1) across cycle classes:

- Period-1 (convergent) channels: mean amplification 0.21
- Period-2 (cycling) channels: mean amplification 0.43

Cycle solutions are roughly $2\times$ better state distinguishers than fixed-point solutions, consistent with their marginal-mode information preservation.

3 Multi-basin dynamics: ergodicity overturned

In a preliminary investigation we reported “100% ergodicity” for D-CTC iteration: every channel tested had a unique fixed point regardless of initial state. That finding was an artifact of restricting the initial-state ensemble to random pure states, which mostly fall in the same basin of attraction.

Corrected test

We probed each channel with 30 *diverse* initial conditions: random pure states, random mixed states (Dirichlet-weighted superpositions of random pure states), basis states, equal-weight superpositions, and the maximally-mixed state. We then clustered the final iterates within Frobenius tolerance 10^{-7} .

Table 2: Number of distinct limit states per channel from 30 diverse inits, $n = 200$ channels per ensemble.

Ensemble	Mean distinct limits	Strict-unique fraction
Haar	1.0	100%
Clifford-like	24.78	18%

A typical Clifford-class channel has approximately 25 distinct limits, depending on initial state. Within a single cycle, however, the iteration is ergodic: from 20 random inits all reaching a particular period-2 channel, both phases of the cycle were visited 10 times each.

Adversarial probing of one channel

For a single Clifford-class channel we tested seven canonical initial states (basis states, pairwise-equal superpositions, and the maximally-mixed state) and observed at least four distinct limit states with purities $\{1.0, 1.0, 0.5, 0.56, \dots\}$. **The same channel produces qualitatively different output states for different input states** — not just different amplifications of the same fixed-point map, but genuinely distinct limits.

4 The encoding continuum

The companion paper [5] distinguishes three encoding regimes:

- **Abstract:** Clifford/Floquet/generic Hamiltonian $\rightarrow$ amplification independent of physical CTC content.
- **Direct-CTC:** matrix entries literally proportional to the spacetime CTC tensor $L_{\text{pair}}(r)$ $\rightarrow$ amplification vanishes at chronology-protected configurations.
- **LP-Hamiltonian:** physics-grounded Hamiltonian on a Clifford backbone $\rightarrow$ pure but quiet (purity 1.0, amplification 0.18).

We test whether this trichotomy is a sharp phase boundary or a continuum by constructing hybrid unitaries

$$U(\delta, t) = \text{polar}[(1 - t)U_{\text{Clifford}} + t \cdot U_{\text{direct-L}}(\delta)] \quad (2)$$

where polar denotes the polar decomposition projecting onto the unitary group. We then sweep $\delta \in [0, 2\pi]$ and measure the ratio of D-CTC amplification at $\delta = \pi$ (chronology-protected) to amplification at $\delta = 0$ (maximum-CTC).

Table 3: Smooth transition from abstract to direct-CTC encoding.

t	Amp at $\delta = 0$	Amp at $\delta = \pi$	Ratio	Regime
0.0	0.268	0.268	1.00	abstract
0.5	0.310	0.140	0.45	threshold
1.0	0.598	0.000	0.00	direct-CTC

The chronology-protection signature emerges gradually: at $t = 0$ the ratio is 1 (no signature), at $t = 0.5$ the ratio drops below 0.5 (signature threshold), and at $t = 1$ the ratio is 0 (full signature). **No sharp phase boundary appears.** The trichotomy of encoding regimes is a continuum parametrised by the channel’s direct-CTC content, and the 50% mixing fraction marks the onset of observable chronology-protection coupling.

5 Re-interpretation of the D-CTC equation

These three findings together suggest a substantial revision of the standard reading of (1). We propose that Deutsch’s self-consistency equation should be re-read as admitting *cycle solutions* of any period $k \geq 1$:

$$\rho_n = \rho_{n+k}, \quad k = 1, 2, 3, 4, \dots \quad (3)$$

with $k = 1$ being the special case of strict fixed points. Cycle solutions of period $k \geq 2$:

- Exist generically in Clifford-class D-CTC channels (66% incidence at $d = 6$);
- Have $\lambda_2(\mathbf{E})$ exactly on the unit circle at angle $2\pi/k$;
- Are quantum-information-preserving (marginal mode);
- Amplify state distinguishability roughly $2\times$ more than period-1 fixed points;
- Live in multi-basin dynamics where initial state selects the particular cycle reached.

The Aaronson–Watrous polynomial-time PSPACE speedup [2] should therefore be re-interpreted as operating on the cycle-solution class, not the fixed-point class. This is consistent with the original theorem (which only requires self-consistency, not strict convergence), and is sharper:

- *The amplification mechanism is marginal-mode preservation*, not fixed-point projection.
- *Implementation requires basin-selection*: the protocol must specify which limit cycle is reached, since multiple are generically available.
- *Numerical practice* should detect cycles, not just fixed points: a tolerance-based fixed-point solver will report “convergence” for a period-2 cycle because consecutive iterates are within tolerance of each other, even though they alternate.

6 Implications

For computational complexity

The Aaronson–Watrous result remains intact, but the channel class realising it shifts: cycle-supporting channels (especially period-2) are the natural site of polynomial-time PSPACE-style amplification. Implementations via post-selection quantum circuits should deliberately target cycle solutions rather than fixed points.

For physics interpretation

If Deutsch’s equation is read literally as a constraint on the quantum state “at the closed timelike curve”, cycle solutions say the state need not be unique — it can be a periodic trajectory through finitely many density matrices. This is a sharper statement than “CTCs admit fixed points”.

For numerical practice

Standard D-CTC simulators that use a fixed-point convergence criterion will report “convergence” for cycle solutions of any period as long as the convergence tolerance exceeds the state-to-state distance within the cycle. We recommend always checking explicitly for cycles by retaining the last $k = 1, 2, \dots$ iterates and testing pairwise distances. The Systrophe library includes such a cycle detector as `detect_cycle_length` in the deep-dive scripts.

7 Reproduction

All findings are reproducible from:

```
python examples/dctc_deep_E1_cycles.py      # cycle distribution
python examples/dctc_deep_E4_trichotomy.py  # smooth transition
python examples/dctc_deep_E8_ergodicity.py   # multi-basin probe
```

at github.com/Zynerji/systrophe v0.17.0 or later.

Acknowledgements

These findings emerged from sustained empirical exploration with the Anthropic Claude coding assistant. The author conceived the three deep-dive experiments, the assistant implemented and executed them, and we jointly interpreted the results. All numerical claims are reproducible from the published scripts.

References

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